A Multivariate Analysis of SDSS Stellar Classification

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1 Introduction

The Sloan Digital Sky Survey (Abolfathi et al., 2018) provides five-band photometric magnitudes for 10,000 celestial objects classified spectroscopically as stars, galaxies, or quasars. This paper addresses two questions: what latent photometric structure distinguishes these object types, and how accurately class membership can be recovered from photometry alone. We apply PCA and Factor Analysis to both raw band magnitudes and derived color indices, with robust outlier detection via Minimum Covariance Determinant estimation.

2 Data

2.1 Dataset Overview

The dataset contains 15 variables, of which 6 are observational metadata (imaging run, camera column, field number, spectroscopic plate, fiber ID, and Modified Julian Date) that reflect the survey's design rather than physical properties of the objects, and are excluded from analysis. Also, two variables that record sky coordinates (right ascension and declination) are omitted from the analysis, as they describe survey coverage but carry no information relevant to object classification. Spectroscopic redshift is also present, and correlates strongly with class: stars ($n = 4,152$) cluster near zero, galaxies ($n = 4,998$) range from near 0 to 0.86, and quasars ($n = 850$) reach values as high as 5.35. Because redshift is derived from the same spectroscopic observation that produces the class label, including it as a predictor would cause label leakage and trivialize the classification task, so it is excluded from modelling. This leaves five photometric band magnitudes capturing ultraviolet, green, red, near-infrared, and infrared (u, g, r, i, z), which will be the primary features used in analysis. These photometric bands are measured on the AB magnitude system, where lower values indicate greater apparent brightness. Though, as we discuss next, these raw magnitudes are not directly comparable across objects due to the confounding effect of distance.

2.2 Color Indices

Since the five photometric magnitudes measure apparent brightness as observed from Earth, objects at greater distances appear fainter regardless of their true physical properties. This introduces distance as a confounding variable where two objects may have identical spectral properties but will have different locations in raw magnitude space. We can control for this through the construction of color indices by taking the difference between adjacent band magnitudes $(u - g, g - r, r - i, i - z)$, which captures the shape of the spectral distribution rather than the magnitudes, and reduces the feature space from five variables to four. In analyses that follow, we compare results on raw bands and color indices, which serve as a robustness check and a test of whether distances affect our findings.

3 Exploratory Data Analysis

3.1 Distributional Characteristics

First, Figure 1A shows the distributions of the five raw photometric bands separated by class. Galaxies and stars follow roughly unimodal distributions across most bands, with substantial overlap particularly in the red, near-infrared, and infrared bands, suggesting these bands alone will not suffice for class separation. Stars exhibit notably wider variance and heavier tails than galaxies in the ultraviolet and green bands, reflecting the wide range of stellar temperatures in the sample. Quasars are more separable across most bands but show the greatest overlap with stars and galaxies in the ultraviolet band, where all three classes occupy similar magnitude ranges. Outliers are visible across all bands, most prominently as isolated extreme values in the near-infrared and green bands. These findings are consistent with the bad photometric measurements flagged in the data documentation (Abolfathi et al., 2018). The impact of these outliers on the covariance structure is quantified in Section 4.

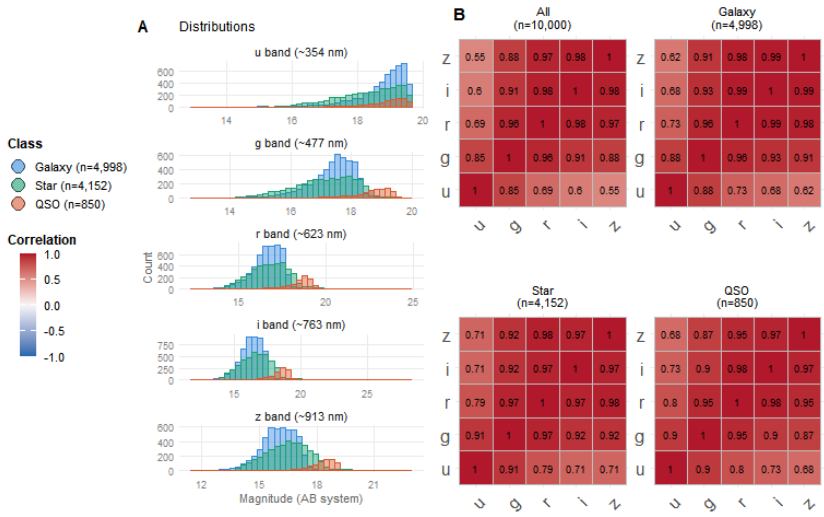

Figure 1: **Raw Photometric Band Distributions and Correlation Structure.** *Panel A:* Overlaid histograms of the five photometric bands (u, g, r, i, z) separated by class. Stars and galaxies overlap substantially in the redder bands, while quasars show the greatest separability in the bluer bands. Extreme outliers are visible across all bands, motivating detection. *Panel B:* Pearson correlation heatmaps for all objects pooled and separated by class. A strong gradient pattern is present globally, with correlations decreasing from $r = 0.98$ between adjacent infrared bands to $r = 0.55$ between ultraviolet and infrared, motivating the use of Factor Analysis.

Controlling for distance through color indices shows more distinct structure, as shown in Figure 2A. The $r - i$ and $i - z$ indices cluster tightly near zero for all classes, with $i - z$ showing near-zero values for stars and slightly positive values for galaxies. This pattern of galaxies having higher color indices than stars is consistent across indices $u - g$ and $g - r$ with a larger range, reflecting that galaxies are typically redder objects as they are dominated by older, cooler stellar populations compared to individual bluer stars. The $u - g$ index looks to be the most discriminating, where quasars cluster around zero, while stars and galaxies occupy the range between one and two. Galaxies show a more balanced bimodal distribution, while there is a small group of stars around the quasar near-zero $u - g$ index. This likely corresponds with a less numerous population of young, hot, blue stars with excess UV, a finding that will be revisited in PCA score plots. The outliers visible in the raw bands appear even more extreme in the color indices, particularly in the otherwise tightly distributed $r - i$ and $i - z$ indices, further motivating formal outlier analysis in Section 4.

3.2 Correlation Structure

Taking a look at Figure 1B shows the Pearson correlation structure between the five raw photometric bands. The weakest correlation occurs between the ultraviolet and infrared bands $r = 0.55$, consistent with their spectral distance, where bands further apart in wavelength share less variance. Conversely, the adjacent near-infrared and infrared bands are nearly perfectly correlated ($r = 0.98$), as expected given their spectral proximity. This pattern of decreasing correlation with increasing spectral distance holds consistently across all band pairs and persists within each class separately as well, indicating this isn't an artifact of mixing objects. The high correlations across all bands motivate the use of Factor Analysis (FA) in future sections to explore latent dimensions.

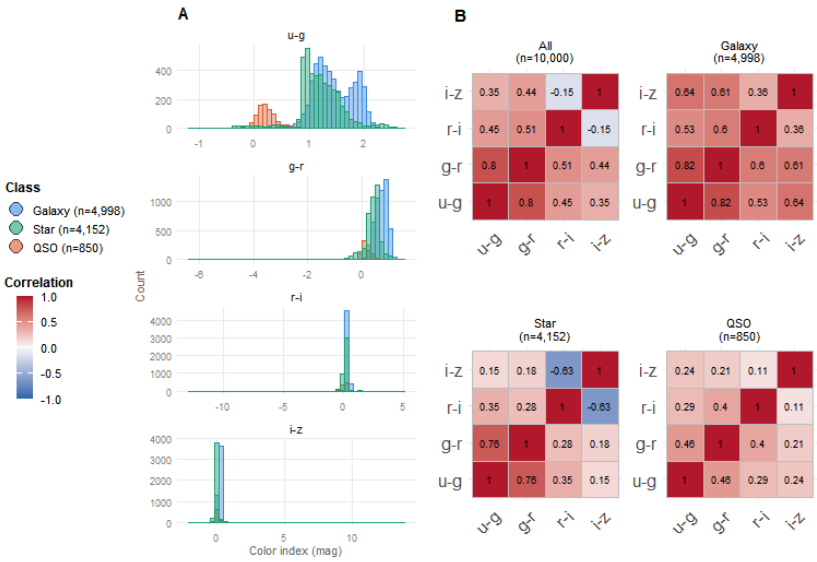

Figure 2: **Color Index Distributions and Correlation Structure.** *Panel A:* Overlaid histograms of the four adjacent color indices ($u-g, g-r, r-i, i-z$) separated by class. $u-g$ is the most discriminating index, with a small subgroup of stars overlapping the quasar distribution, likely hot young blue stars with UV excess. Outliers are most severe in $r-i$ and $i-z$, motivating Section 4. *Panel B:* Pearson correlation heatmaps pooled and by class. The pooled $r-i$ vs. $i-z$ correlation ($r = -0.15$) masks a Simpson's Paradox: $r = -0.63$ within stars but $r = 0.36$ within galaxies, motivating class-conditional analysis.

Figure 2 shows a similar, but weaker correlation structure among the four color indices than that found in the raw photometric bands (Figure 1). The strongest correlation is between adjacent indices $u-g$ and $g-r$ ($r = 0.80$), followed by $g-r$ and $r-i$ ($r = 0.51$), consistent with the pattern of nearby indices sharing more variance. However, $r-i$ and $i-z$ break this pattern with a weak negative pooled correlation ($r = -0.15$). Separating the heatmaps by class shows this is a case of Simpson's Paradox. The source of negative pooled correlation comes from within the stars class, where $r-i$ and $i-z$ are moderately negatively correlated ($r = -0.63$). Within galaxies however, the same adjacent pair is positively correlated ($r = 0.36$), and the most distant indices are more strongly correlated ($r = 0.64$), likely due to the makeup of galaxies being more red. Quasars exhibit notably weaker correlations across all color index pairs (all values below $r = 0.50$), consistent with them occupying a more distinct region of color space, as seen in the distributional analysis (Section 3.1). This further supports the use of class-conditional analysis throughout subsequent analyses.

4 Preprocessing

As observed in the distributional histograms (Figure 1A and 2A) and noted in the data documentation, extreme outliers are present across all photometric bands, with the most severe cases concentrated in the $r - i$ and $i - z$ color indices. Since outliers distort the covariance structure underlying both Factor Analysis (FA) and Principal Component Analysis (PCA), we systematically detect and remove them using the Minimum Covariance Determinant (MCD) estimator, visualized through Distance-Distance (D-D) plots that compare the classical and robust Mahalanobis distances.

We consider two thresholds at which to classify outliers. First, the conventional 95% MCD, which corresponds to a χ^2 cutoff at the 97.5% percentile and flags observations whose robust distance exceeds $\sqrt{\chi^2_{p,0.975}}$. However, this threshold was found to exclude too many non-outlier observations. Instead, we choose our primary threshold to be within five standard deviations (5SD), following the convention in particle physics for declaring observations physically impossible (Lyons, 2013). At this threshold, only genuinely anomalous measurements are removed while preserving exceptional naturally occurring cases. Results under 95% MCD are included as a sensitivity check in the classification accuracy results (Figure 6A).

4.1 Pooled vs. Within-Class MCD

Applying MCD to the 95% pooled dataset showed that 94% of quasar observations are flagged as outliers under the color index feature space and 89.6% for raw photometric bands. Rather than being a result of measurement errors, this reflects the physical separability of quasars established in the distributional analysis, where quasars occupy a distinct region on the color bands: g, r, i, z (Figure 1), and more clearly for color indices: $u - g$ and $g - r$ (Figure 2). As such, quasars appear as outliers relative to the much larger star and galaxy populations. Thus, we opt to employ outlier detection within classes to better account for the distribution of the data.

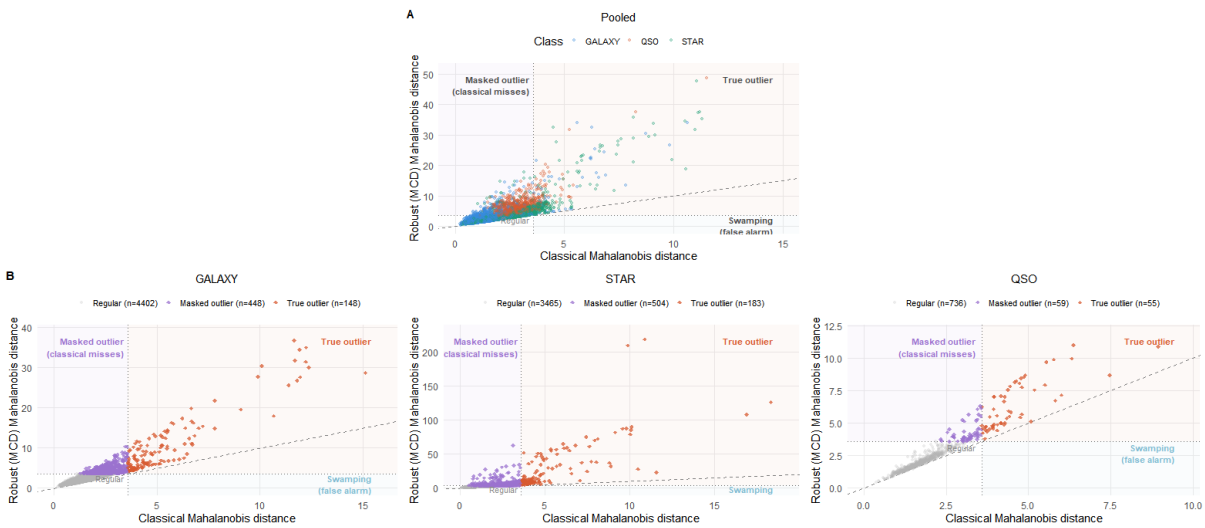

Figure 3: **Distance-Distance Plots for Pooled and Within-Class MCD Outlier Detection.** *Panel A:* Pooled MCD D-D plots for raw bands and color indices. QSOs dominate the outlier quadrants (94% flagged for color indices), motivating within-class detection. *Panel B:* Within-class MCD D-D plots. Different axis scales across classes corroborate the distributional differences between object types and reduce the total outlier rate from 17.7% to 13.7% for color indices, with removal now distributed more evenly across classes.

To avoid removing genuine quasar observations, we perform MCD within each class separately. This reduces the total outlier rate for a 95% threshold from 17 % pooled to 14% within-class for raw photometric bands, and 17.7% to 13.7% for color indices, with outliers now distributed more evenly across classes: 11% of galaxies, 12% of quasars, and 17.2% of stars being flagged. The class-separated D-D plots (Figure 3B) show dramatically different axis scales across classes, corroborating our observational differences between stellar object distributions and further supporting the need for class-conditional rather than pooled outlier estimation.

4.2 Effect on Feature Structure

The impact of outlier removal on downstream analyses is most clearly seen in Kaiser-Meyers-Olkin (KMO) sampling adequacy scores. For raw photometric bands, the overall KMO decreases slightly from 0.79 to 0.73 under both 2SD and 5SD outlier removal, supporting our choice to remove them as they inflated apparent shared variance across bands. For color indices however, the KMO improves substantially from 0.58 to 0.75, crossing the threshold from "miserable" to "meritorious" for FA (Kaiser, 1974). Thus, outlier removal allows for more appropriate method selection and is the main justification for the preprocessing step.

5 Dimension Reduction

5.1 Principal Component Analysis

To uncover the latent photometric structure of the dataset, we apply PCA to both the raw photometric bands and the color indices, comparing the two feature representations and examining the impact of outlier removal on the recovered dimensions. All predictors were normalized and centered prior to analysis, since the five bands and four color indices are measured on different scales, and both PCA and FA are sensitive to variance differences across variables.

5.1.1 Model Specification

We first justify retaining two principal components (PC) despite the criteria suggesting one would suffice. Looking at the scree plot in Figure 4A, PC1 is the only component satisfying the Kaiser Criterion across both feature representations, with eigenvalues $\lambda_1 = 4.41$ and $\lambda_2 = 3.35$ for raw bands and color indices respectively after 5SD outlier removal.

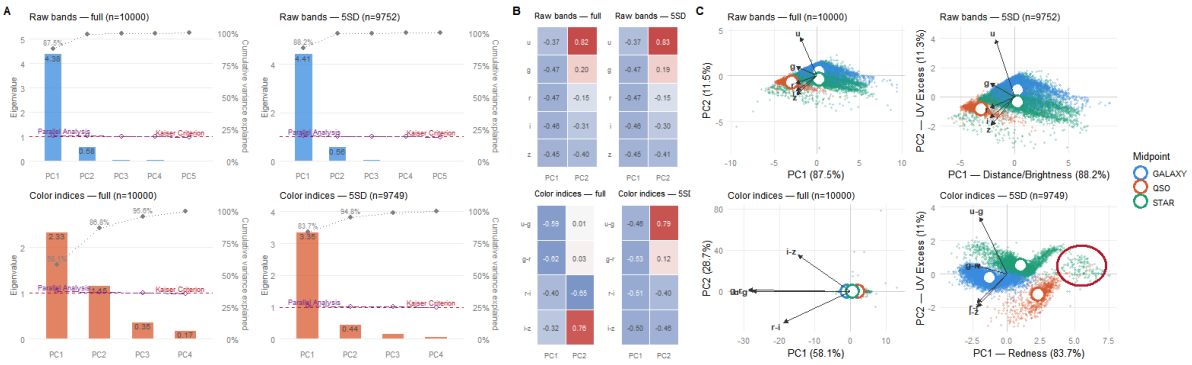

Figure 4: **PCA Scree Plots, Loading Heatmaps, and Score Biplots.** *Panel A:* Scree plots for full and 5SD datasets across both feature representations. Criterion suggests only one component be retained, though two are for interpretability. *Panel B:* Varimax loading heatmaps. PC1 captures overall brightness in raw bands and a redness gradient in color indices; PC2 captures UV excess in both representations. *Panel C:* Score biplots with class centroids and loading arrows. QSOs are separable primarily along PC2, with a circled subgroup of stars overlapping the quasar region in the color index space.

The eigenvalue ratios are large, with approximately 7.9:1 ($\lambda_1/\lambda_2 = 4.41/0.56$) for raw bands and 7.6:1 ($\lambda_1/\lambda_2 = 3.35/0.44$) for color indices, indicating that the first principal component dominates the variance structure. Parallel analysis corroborates this, with only PC1 exceeding the 95th percentile of simulated null eigenvalues in both feature sets.

That said, PC1 alone accounts for approximately 88% of variance in raw bands and 84% in color indices, this is above the 80% threshold, however some use 90%, and this motivates our decision to retain a second principal component for better interpretability.

It is worth noting that in the full color indices dataset (prior to 5SD outliers removal), PC2 appears substantially more prominent with an eigenvalue of $\lambda_2 = 1.44$, exceeding the Kaiser Criterion threshold and accounting for an additional 36% of variance. However, the corresponding biplot shows that this inflation is driven by extreme outliers loading disproportionately on the second component. The drop to $\lambda_2 = 0.44$ confirms the necessity of removing extreme outliers.

5.1.2 Interpretation

Throughout this section, axis labels denoting physical interpretations (e.g. "Distance/Brightness", "UV Excess") are shown only for the 5SD outlier-removed panels, as the full dataset dimensions are partially distorted by extreme measurements and do not support stable labelling.

We first look at the loading heatmaps and biplots in the top row of Figure 4(B and C) for the raw photometric bands. PC1 has moderate negative loadings on all five bands (-0.37 to -0.47), meaning it is overall mostly a brightness axis, so it largely reflects distance/luminosity rather than color. The biplot confirms this, with the galaxy and star centroid sitting at high PC1 (faint and distant), while quasars are low on PC1. This further supports our introduction of color indices to control for this confounding variable.

PC2 separates the blue/UV from the red/infrared end, with u and g loading positively (0.83 and 0.19) against the negative loadings of r, i, z (-0.15 to -0.41). Thus, objects with low PC2, such as quasars, have relatively brighter UV compared to their infrared counterparts. Stars are also positioned lower on PC2 than galaxies, since as mentioned before, galaxies are typically made of older redder stars compared to younger blue ones.

Next, examining the color indices (bottom row of Figure 4B and 4C), PC1 loads with a similar negative pattern across the board (-0.46 to -0.53) to that of the raw photometric bands. Since distance is controlled this time by taking the difference between bands, PC1 is more of a blue-to-red color gradient axis. This once again agrees with the knowledge that galaxies are older and thus redder and lower, with the most negative midpoint of the classes.

PC2 again reflects a blue versus red color contrast: in the 5SD dataset, $u - g$ loads strongly positively (0.79) while $r - i$ and $i - z$ load negatively (-0.40 and -0.46), with $g - r$ near zero (0.12). Objects with high PC2 therefore, have anomalously blue $u - g$ colors relative to their redder indices, pointing to UV excess as opposed to UV in general from raw photometric bands and more accurately describing the photometric properties of lower PC2 scoring quasars. Notably, there is a sign flip in the loadings of PC2 between the full dataset and 5SD, another consequence of the extreme $r - i$ and $i - z$ identified.

5.2 Factor Analysis

5.2.1 Assumption Testing

To first confirm that our dataset is best suited for FA, we test some assumptions. First, we note that the KMO measure was taken across all dataset variants, and after outlier removal, both raw photometric bands (0.73) and color indices (0.75) are considered "meritorious" under Kaiser's rule of thumb (Kaiser, 1974). This is consistent with our global Pearson correlation maps, where correlations ranged from $r = 0.98$ to $r = 0.55$. Finally, Bartlett's test of sphericity strongly rejects $p \approx 0$ that the null correlation matrix is an identity matrix, confirming there is

meaningful correlation structure (Bartlett, 1951). Having established that the data satisfy the necessary assumptions, we may use FA to find underlying latent dimensions.

To further support our selection of FA to model the data, we must confirm the multivariate normality assumption. Mardia's test is applied to a random subsample of $n = 1,000$ observations to mitigate the well-known rejection artefact that occurs at large sample sizes, where even trivial departures from normality produce significant results due to statistical power alone. Even on the subsample, skewness and kurtosis fail ($p \approx 0$), however this is consistent with the mixture structure of the data, as pooling three photometrically distinct classes produces a multivariate distribution that is not jointly Gaussian, regardless of within-class normality.

To better support multivariate normality, we use a test more robust to large sample sizes. Negentropy is computed for each indicator, where values range from near zero to < 0.01 across all indicators, confirming Gaussianity as all values are near zero. These results provide us with sufficient evidence to proceed under the assumption of multivariate normality. This further supports our choice of FA, which assumes Gaussian latent factors, in contrast to Independent Component Analysis (ICA), which relies on non-Gaussianity for identifying the signal.

5.2.2 Model Specification

We first justify the choice of Maximum Likelihood (ML) estimation over Principal Axis Factoring (PAF). While PAF is more robust to multivariate normality violations, ML is preferred in this setting as we established that multivariate normality holds during assumption testing through near-zero negentropy values (< 0.01).

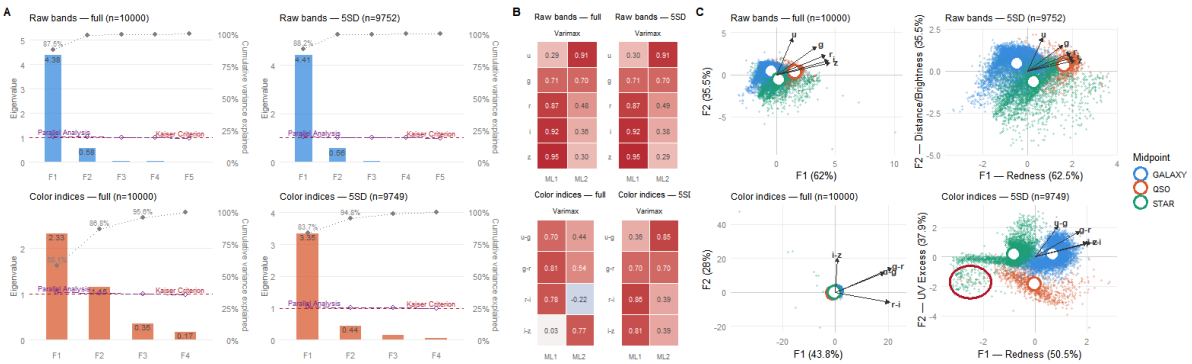

Figure 5: **FA Scree Plots, Varimax Loading Heatmaps, and Score Biplots.** *Panel A:* Scree plots for full and 5SD datasets across both feature representations. Similar to PCA, two factors are retained for interpretability. *Panel B:* Varimax loading heatmaps. ML1 captures red light emission in raw bands and a redness gradient in color indices; ML2 captures a monotonically decreasing UV contribution in raw bands and a UV-excess axis in color indices. *Panel C:* Score biplots with class centroids and loading arrows. QSO separability is driven by ML2, with a circled subgroup of stars overlapping the quasar region in the color index space, corresponding to the hot young stars discussed in Section 7.1.

To determine the number of factors to retain, we used parallel analysis, alongside the scree plot in Figure 5A, which shows various criteria. The scree plots show a sharp elbow after the first factor, immediately suggesting that variance is dominated by a single factor. Similarly to the scree plots for PCA (Figure 4A), the eigenvalue ratio favors a single-factor solution: for raw photometric bands $\lambda_1/\lambda_2 = 4.41/0.56 \approx 7.9$ and for color indices $\lambda_1/_2 = 3.35/0.44 \approx 7.6$. The Kaiser criterion and parallel analysis agree, with only λ_1 exceeding both the threshold of 1 and the 95th percentile of simulated null eigenvalues in all dataset variants. Cumulative variance again is the sole criterion that falls marginally short, reaching 88.2% and 83.7% for raw photometric bands and color indices respectively after one factor. We therefore retain a second factor for interpretability and to capture more variance in the data, as a single factor

would capture only a general brightness or redness signal without distinguishing the photometric contrast between objects.

When selecting for rotation, we opt for varimax rotation since, while promax produces sharper loading contrasts, the oblique solution yields a factor correlation of approximately 0.70 between the two factors. Such a high correlation implies that the factors are not meaningfully distinct constructs, undermining the interpretability gain that oblique rotation is intended to provide. Thus, varimax's orthogonality constraint is more appropriate here, as the two retained factors will represent separable photometric dimensions as shown in the score biplots.

5.2.3 Interpretation

Before interpreting loadings, note again that due to the AB magnitude system, the relationship between magnitude and brightness is inverted, so lower values correspond to brighter objects. As a result, factor score directions should be interpreted with this inversion in mind.

First, the varimax loading heatmaps for photometric bands (Figure 5B top row) show ML1 carrying positive loadings across all bands, focusing primarily on the stronger positive correlations between bands green (g) (0.71), increasing monotonically to infrared (z). This factor represents red light emission, explained by galaxies being lowest on the ML1 scale as the relationship is inverse, while quasars with excess UV light score highly. ML2 loads positively again, though monotonically decreasing from ultraviolet (u) (0.91) to infrared (z) (0.29). ML2 may be interpreted as the overall brightness indicator, hinted by the soft asymptotic barrier present around 2 on factor 2. We will see how this changes once we account for the differences between bands in color indices.

For color indices (Figure 5 bottom row), ML1 loads positively across all indices, with the strongest contributions from the central indix $r - i$ (0.86), then $i - z$ (0.81), followed by $g - r$ (0.70), and a comparatively weaker loading on $u - g$ (0.36). This factor can be interpreted as a redness or temperature axis, where galaxies tend to score higher on ML1 while individual stars fall lower along this dimension. In contrast, ML2 reflects an excess UV axis, with stronger loadings on shorter wavelength indices (0.85 for $u-g$ and 0.70 for $g-r$), and weaker contributions from the longer wavelength indices (0.39). This helps explain why objects with strong UV emission, such as quasars and a specific sub-group of stars discussed later, tend to score strongly negative along this factor.

5.3 Comparing PCA and FA

In the raw photometric bands, both PCA and FA identified a dominant first dimension associated with overall brightness brought about by the influence of distance. However, once color indices are introduced, both methods shift toward capturing spectral properties. For color indices, the first dimension of both PCA and FA aligned with a red-blue or temperature gradient, separating older, redder galaxies from blue stellar populations.

The second dimension also showed strong agreement across methods. In PCA, PC2 distinguishes ultraviolet from infrared contributions, while in FA, ML2 is more cleanly interpretable as a UV-excess factor due to the rotated structure. This highlights a key difference between the methods, where PCA mixes multiple interpretations, whereas FA produces more interpretable latent constructs. That being said, both methods biplots captured the significant difference between quasars and other objects.

Common to both approaches was the influence of outliers on the second dimension. Before 5SD removal, both PCA and FA show an inflated second component/factor, particularly for color indices. This resulted in artificially large eigenvalues for PC2 and very distorted score biplots.

6 Object Classification

6.1 Method Selection

Next, to classify the celestial objects into three classes: galaxies, stars, and quasars, we apply Linear Discriminant Analysis (LDA) and multinomial logistic regression. Both Methods appear to be well-suited to our data since from the distributional analysis and PCA/FA score biplots, natural groupings were quite explicit. LDA additionally assumes equal within-class covariance matrices and approximate multivariate normality, which we have found during assumption testing with near-zero negentropy. Logistic regression has no distributional assumptions, making it a useful comparison to LDA.

We deliberately exclude kernel methods, despite the slight non-linear structure visible for stars in the color index biplots. We will find that LDA and logistic regression already achieves strong classification performance, so the marginal accuracy gain from kernelising would come at the cost of interpretability. Regularisation methods such as ridge or lasso are similarly unnecessary here, as with only five photometric bands or four color indices as predictors, the dimensionality is far too low for overfitting to be a concern.

6.2 Cross-Validation Design

Ten-fold cross-validation is used to evaluate both classifiers, with folds constructed using both stratified and non-stratified sampling to measure the effects of maintaining the natural class imbalance of the dataset (50% Galaxy, 41.5% Star, 8.5% QSO).

6.3 Results

Both LDA and logistic regression achieve strong classification performance across all dataset variants (Figure 6A), with logistic regression consistently outperforming LDA by approximately 2% in overall accuracy. The performance gap is most pronounced in QSO recall, where logistic regression correctly identifies 92.7% and 91.5% of quasars for raw bands and color indices respectively, while LDA achieves only 84.8% and 84.6% on the same categories.

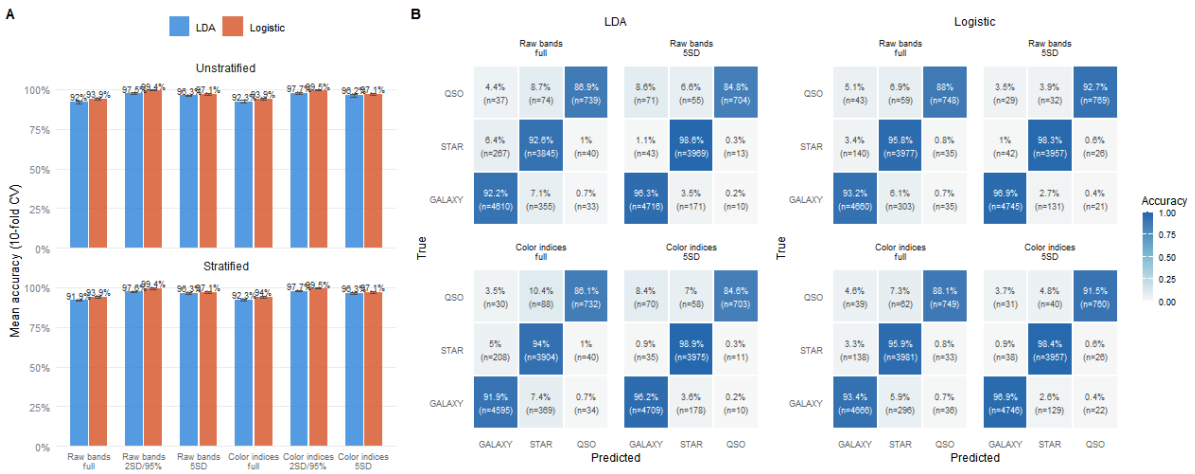

Figure 6: **Classification Accuracy and Confusion Matrices for LDA and Logistic Regression.** *Panel A:* 10-fold CV classification accuracy. Logistic regression consistently outperforms LDA by ∼2%, with stratified and unstratified folds yielding nearly identical results. *Panel B:* Confusion matrices for both classifiers. The highest misclassification rate is quasars confused with stars, while the most numerous errors are stars misclassified as galaxies.

Stratified and unstratified cross-validation yield nearly identical mean accuracy, with stratification producing slightly lower variance across folds, hence class imbalance is mild enough

not to cause severe instability. The 2SD outlier-removed datasets, visible in Figure 6A, show near-perfect logistic regression accuracy of 99.4%. While an expected result, it shows that the 2SD threshold removes too many naturally occurring extremes, inaccurately reflecting the full challenge of classifying celestial objects.

Examining the confusion matrices (Figure 6B), the most prevalent misclassification pattern is stars being predicted pattern is stars being predicted as galaxies. This is consistent with the EDA finding that stars and galaxies overlap substantially across most photometric bands. Meanwhile, the highest misclassification rate comes from quasars being confused with stars, reflected by the partial overlap in UV-excess space identified in PCA and FA biplots, and the separability found in $u - g$ color index during distributional analysis.

7 Discussion

7.1 Sub-Groups of Stars

We alluded throughout the analysis to a subpopulation of stars with anomalously low $u-g$ color indices, first in the distributional analysis, then recurring as the circled cluster in the FA and PCA score biplots for color indices (Figure 4C and 5C). Hypothesizing that this subgroup of stars would cause trouble in linear classification, we visualized the incorrect predictions from our best model, logistic regression, in Figure 7. Here, we show that there is only 70.1% prediction accuracy inside the circled region, reflecting the failure of a linear photometric classification in this dataset.

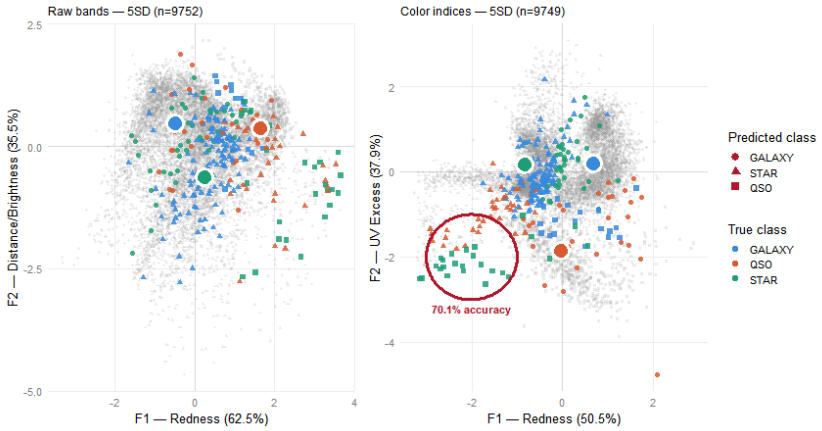

Figure 7: **FA Score Biplots with Logistic Regression Prediction Correctness.** Incorrect predictions are coloured by true class and shaped by predicted class, with correct predictions shown in grey. The circled region in the color index space achieves only 70.1% accuracy, isolating the hot young star subgroup whose UV excess overlaps with that of low-redshift quasars, representing the main failure of linear photometric classification.

This subgroup of stars is most likely hot, young O, B, and A-type stars whose surface temperatures (10,000-50,000 K) produce strong UV through blackbody radiation, placing them in the same region of $u - g$ color space as low-redshift quasars (Martins et al., 2005).

7.2 Convergent Evidence

Several findings were independently confirmed across multiple methods, which lend confidence to the core conclusions of this analysis.

The role of the $u - g$ color index as the primary discriminator of quasars was independently supported across the distributional analysis (Section 3.1), the PCA and FA biplots (Figures 4C and 5C), and the classification confusion matrices (Figure 6B). Each method, from different

analytical angles, pointed to the same conclusion: quasars are separable primarily through their UV excess, with star-quasar confusion concentrated where this UV signal overlaps with that of hot and young O, B, and A-type stars.

The necessity of outlier removal was also validated across multiple analyses. The inflation of PC2 and ML2 eigenvalues before 5SD removal (visible in Figures 4A and 5A), the KMO improvement for colr indices from 0.58 to 0.75, and the near-perfect but unreliable 99.4% accuracy under 2SD all independently support the same conclusion: the extreme photometric measurements identified are genuine artefacts that distort dimension reduction and classification.

7.3 Divergent Evidence

Despite these agreements, there are several points of divergence showcasing the complexity of the data that a single method alone would have missed.

Most notably, PC1 for PCA in the raw photometric bands ended up being largely an overall brightness indicator confounded by distance, while ML1 for FA serves as a measure of redness and acts as a separator for stars and galaxies. This divergence is resolved by color indices, where both methods align on redness. Additionally, PCA and FA produce subtly different second dimensions in the raw band space: PC2 mixes UV brightness with infrared contrast, while FA's varimax rotation yielded a cleaner monotone UV-excess factor. The FA result is more interpretable, but the PCA result is more faithful to the raw variance decomposition.

7.4 Limitations

A recurring interpretive caveat throughout this analysis is the inverted relationship between magnitude and brightness under the AB system, where lower values correspond to brighter objects. As a result, factor and component score directions are reversed relative to intuitive brightness expectations: e.g., a high PC1 or ML1 score indicates a fainter, not brighter, object. While this inversion is accounted for in all interpretations provided, it introduces a layer of complexity that makes the results less immediately accessible to readers unfamiliar with astronomical magnitude systems.

Several limitations constrained the conclusions of this analysis. The 5SD outlier threshold, while physically motivated by particle physics conventions (Lyons, 2013), remains arbitrary, thus a different threshold would yield different retained samples, as the sensitivity check in the appendix for 2SD demonstrates.

Additionally, the linear classifiers and linear dimension reduction methods used here cannot fully capture the non-linear boundaries visible in the score biplots, particularly at the star-quasar boundary. Future studies may want to explore methods such as k-means of a Gaussian Mixture Model, which may be better suited to formally identify subgroups within the star class (e.g. the hot blue young stars and the cooler red older stars).

Finally, the analysis treats raw photometric bands and color indices as separate feature sets. Combining them into a joint feature space may yield richer structure, at the cost of reintroducing the distance confound.

8 Conclusion

Factor Analysis and PCA applied to SDSS photometric data reveal a consistent two-dimensional structure across both raw bands and color indices: a dominant brightness or redness axis driven by distance and stellar population age, and a secondary UV-excess axis that separates quasars from other objects. Logistic regression achieves strong classification performance on this structure, outperforming LDA by approximately 2%, with the primary failure mode concentrated in a subgroup of hot O-, B-, and A-type stars whose UV excess overlaps with that of low-redshift quasars. Future work may explore unsupervised clustering methods to formally characterize

stellar subpopulations, and a joint feature space combining raw bands and color indices to capture both distance-dependent and intrinsic spectral structure simultaneously.

AI Declaration

AI assisted with grammar in the report and generated code to perform analyses.

References

Abolfathi, B., Aguado, D. S., Aguilar, G., et al. (2018). The fourteenth data release of the sloan digital sky survey. *The Astrophysical Journal Supplement Series*, 235(2):42.

Bartlett, M. S. (1951). The effect of standardization on a chi-square approximation in factor analysis. *Biometrika*, 38:337–344.

Bernaards, C. A. and Jennrich, R. I. (2005). Gradient projection algorithms and software for arbitrary rotation criteria in factor analysis. *Educational and Psychological Measurement*, 65:676–696.

Kaiser, H. F. (1974). An index of factorial simplicity. *Psychometrika*, 39:31–36.

Kassambara, A. (2023). *ggcorrplot: Visualization of a Correlation Matrix using 'ggplot2'*. R package version 0.1.4.1.

Korkmaz, S., Goksuluk, D., and Zararsiz, G. (2014). Mvn: An r package for assessing multivariate normality. *The R Journal*, 6(2):151–162.

Lyons, L. (2013). Discovering the significance of 5 sigma.

Maechler, M., Rousseeuw, P., Croux, C., Todorov, V., Ruckstuhl, A., Salibian-Barrera, M., Verbeke, T., Koller, M., Conceicao, E. L. T., and Anna di Palma, M. (2026). *robustbase: Basic Robust Statistics*. R package version 0.99-7.

Martins, F., Schaerer, D., and Hillier, D. J. (2005). A new calibration of stellar parameters of Galactic O stars. *Astronomy & Astrophysics*, 436:1049–1065.

Pedersen, T. L. (2024). *patchwork: The Composer of Plots*. R package version 1.3.0.

Venables, W. N. and Ripley, B. D. (2002). *Modern Applied Statistics with S*. Springer, New York, fourth edition. ISBN 0-387-95457-0.

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., and Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43):1686.

Wilke, C. O. (2024). *cowplot: Streamlined Plot Theme and Plot Annotations for 'ggplot2'*. R package version 1.1.3.

William Revelle (2026). *psych: Procedures for Psychological, Psychometric, and Personality Research*. Northwestern University, Evanston, Illinois. R package version 2.6.1.